



Comparison of the microstructures and mechanical properties of Ti–6Al–4V fabricated by selective laser melting and electron beam melting

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ABSTRACT

The microstructure and mechanical properties of SLM and EBM Ti–6Al–4V samples have been compared. The effect of part size and orientation on the defects, microstructure and their contribution to the tensile and fatigue properties were elucidated. As-fabricated SLM and EBM samples mainly consisted of α' and $\alpha + \beta$ phases, respectively. Pores were the main defects in SLM and EBM samples, and closely related to scanning strategies and energy input. The porosity of SLM samples was higher compared to EBM samples. The part size had an obvious influence on the microstructure and mechanical properties of EBM samples but less so for SLM samples. Both SLM and EBM samples possessed higher strength and better ductility in the vertical orientation than those in the horizontal orientation. The tensile strength of SLM samples was significantly greater than that of EBM samples whereas the ductility was much lower. Due to the pores contained in samples, fatigue strength of both EBM and SLM samples was lower than those of cast and annealed alloys. However, hot isostatic pressing (HIP) significantly increased the fatigue limits of both SLM and EBM samples to above 550 MPa by closing of the pores.

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1. Introduction

Additive manufacturing (AM), also called rapid prototyping and rapid manufacturing, is a process of adding material to make objects from 3D model data, as opposed to subtractive manufacturing methodologies [1]. The AM is an emerging advanced manufacturing technique that creates parts in point-wise matter. In both Selective Laser Melting (SLM) and Electron Beam Melting (EBM) process, metal powder is selectively melted to form each layer. In SLM, the heat source is a laser, while in EBM it is an electron beam. Both techniques have been attracting increasing attention in aerospace and biomedical fields due to their advantages of producing prototypes or finalized parts rapidly and cost-effectively, whilst providing accurate control over both internal architectures and complex-shapes. Many materials, such as stainless steel 316, cobalt-based, titanium alloys, and aluminum alloys have been fabricated using EBM and SLM techniques and their microstructure, mechanical properties and potential applications have been reported [2–6].

Ti–6Al–4V is the most widely used $\alpha + \beta$ dual-phase titanium alloy. Due to its broad applications in biomedical devices, aerospace, marine,

and offshore applications, Ti–6Al–4V parts fabricated by SLM and EBM have been investigated extensively [7–19]. Most of investigations have been focused on developing a comprehensive understanding of the processing-microstructure-properties relationships. For the EBM technique, the as-fabricated Ti–6Al–4V parts generally consists of columnar prior β grains (formed in the β phase field at high temperature) delineated by grain boundary α and a transformed $\alpha + \beta$ structure within the prior β grains [9]. The mechanical properties have been reported to be comparable to the wrought material [10,11]. However, the microstructure and mechanical properties have been reported to be closely related to the processing parameters, part orientation, location and post heat treatment [12,13]. For example, variation of build temperature has a significant effect on the properties and microstructure of as-deposited samples [12]. Some works reported that the orientation has no effect on ultimate strength (UTS) or yield strength (YS), it has a large influence on elongation [12]. While others reported that the UTS for the horizontally built samples was marginally higher compared to the vertically built counterpart [20,21]. The prior β grain size, α lath thickness and mechanical properties, including microhardness, were not found to vary as a function of distance from the build plate [13]. The tensile properties and elongation have been shown to be largely independent of the powder size (25 μm ~ 100 μm) or layer thickness (70 μm and 50 μm , respectively), whereas the surface appearance was

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found to be different for the samples fabricated with different powder sizes [14].

As for SLM Ti–6Al–4V parts, the as-fabricated microstructure is dominated by columnar β grains and α' martensite due to the fast cooling rate [15]. The microstructure, roughness, densification and microhardness of Ti–6Al–4V parts were also closely related to the processing parameters. A fully dense Ti–6Al–4V part with high microhardness and smooth surface can be manufactured by adjusting the power and scanning speed of laser, and the density can be comparable to that of bulk Ti–6Al–4V alloy [16]. The porosity level generally decreased with increasing laser power and/or decreasing laser scanning speed [15]. Because of the hard martensite, the as-fabricated SLM products show very high tensile strengths but poor ductility [15]. It was reported that the fatigue life of SLM product was lower than that of wrought material, which was attributed to microstructure, porosity, surface finish and residual stress within the part [17,18]. Hot isostatic pressing (HIP) considerably improved the fatigue strength such that properties comparable to wrought alloys could be produced [18,19]. The HIP process also increased the ductility but led to a reduction in strength [18,19].

Although many studies have been carried out on the microstructural and mechanical properties of EBM and SLM Ti–6Al–4V samples respectively, little attention has been paid to a comparison between these two processes [11,22]. Therefore, this work is aimed at comparing EBM and SLM processes in terms of microstructure, tensile properties, and fatigue properties of Ti–6Al–4V alloys. The effect of part size, orientation on the defect, microstructure and their contribution to the tensile and fatigue properties were elucidated.

2. Experimental

2.1. Materials

The medical grade Ti–6Al–4V ELI (Extra Low Interstitial) powder was used for both the EBM and SLM process. The Ti–6Al–4V powders for EBM and SLM are provided by Arcam AB (Sweden) and LPW Technology (Cheshire, UK), respectively. The morphology of the powders is shown in Fig. 1(a) and (b), indicating the powders are nearly spherical. The particle size of the powders is measured by laser scattering particle size distribution analyzer (HORIBA LA-920). The results indicated that the average particle size for EBM and SLM is 77 μm (d_{10} : 51 μm , d_{50} : 72 μm , d_{90} : 109 μm) and 35 μm (d_{10} : 26 μm , d_{50} : 38 μm , d_{90} : 51 μm) respectively. The chemical compositions obtained by wet chemical and gas analyses of Ti–6Al–4V powders are shown Table 1.

Ti–6Al–4V samples were produced using an Arcam A1 EBM machine or a Realizer SLM 100. In the EBM process, a voltage of 60 kV and electron beam size of $\sim 200 \mu\text{m}$ were used and the process was kept under vacuum at 10^{-3} mbar, controlled by using helium as regulating gas. The process started with preheating the powder prior to melting. For each layer, a pre-scan was performed which scanned the powder bed 11 times with a beam current of 30 mA and a scan velocity of 15,000 mm/s. This heated the powder layer to 1003 K. Following

Table 1

Chemical compositions of Ti–6Al–4V ELI powders, and as-fabricated parts of SLM and EBM samples (in weight %).

		Ti	Al	V	O
Powders	EBM	Bal.	6.40	4.05	0.13
	SLM	Bal.	6.25	4.04	0.14
As-fabricated	EBM	Bal.	5.97	4.16	0.15
	SLM	Bal.	5.70	3.85	0.20

preheating, samples were produced with a speed function of 39 (scan current of 19 mA, scan speed of 1500 mm/s), scan spacing of 0.2 mm, and layer thickness of 0.05 mm. In the SLM process, samples were produced with a laser spot size of $\sim 40 \mu\text{m}$, scan speed of 1000 mm/s, a laser power of 200 W, scan spacing of 0.1 mm, and layer thickness of 0.05 mm. According to method introduced in [23], the calculated energy density input for SLM and EBM is 40 J/mm³ and 76 J/mm³, respectively. Parts were built in 3 mm stripes using an alternating scanning strategy (that is the scanning vectors were rotated 90° between layers) without overlap. Parts were built on a CP-Ti substrate which had been heated to 473 K. Prior to building, the chamber was purged with high purity Ar until the oxygen content was <200 ppm.

The Ti–6Al–4V rods with different orientations (Vertical-V, Horizontal-H) and different diameters (1.2 mm, 4 mm, 7 mm) were fabricated by SLM and EBM, and then been machined to dimensions prior to testing. The details and abbreviations are shown in Fig. 2(a) and Table 2. Some of the SLM-V7 and EBM-V7 samples were HIPed on a QIH-15 hot isostatic pressing furnace at a temperature of 1203 K with pressure of 130 MPa for 3 h (18 ks). The chemical compositions of as-fabricated Ti–6Al–4V SLM and EBM samples are shown Table 1. The slight decrease of Al and increase of O in both EBM and SLM samples may not significantly influence their morphology and phase [24].

2.2. Microstructural analysis

Microstructural analysis was performed using optical microscopy (OM) (ZEISS-AXIO), SEM (JSM-6510A), and transmission electron microscopy (TEM) (Philips EM420). The specimens for the OM analysis were mechanically polished and then etched in a solution consisting of 2 vol.% HF, 5 vol.% HNO₃, and 43 vol.% H₂O. The Ti–6Al–4V powders, surface of EBM and SLM samples, and the fracture surface of the tensile and fatigue samples are observed by SEM (secondary electron mode). Before observation, they were cleaned by acetone, alcohol, and water, respectively. The TEM specimens were prepared by electrolytic thinning in a solution with 60 ml perchloric acid, 85 ml n-butanol and 150 ml methanol at ~ 240 K. The grain size was measured using the linear intercept method, and more than 50 grains are measured for each sample. Phase constitutions were examined on a Bruker D8 Discover 2D X-ray diffractometer (XRD) using a Cu-K α radiation source with an accelerating voltage of 40 kV and a current of 250 mA.

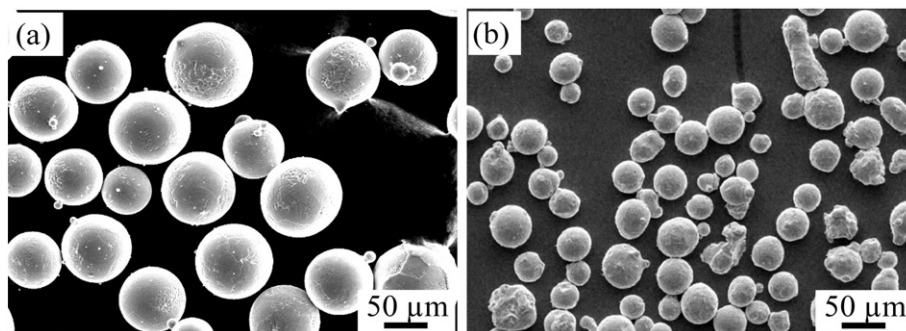


Fig. 1. Scanning electron microscopy (SEM) images of the Ti–6Al–4V ELI powders used for (a) EBM and (b) SLM.

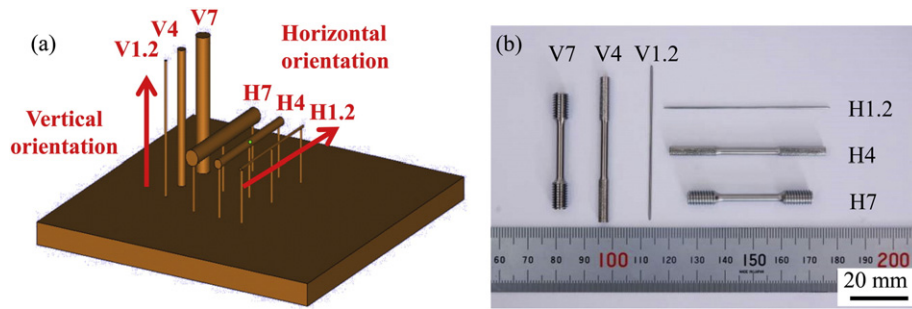


Fig. 2. (a) CAD models of the builds in this work. (b) Machined tensile specimens with nomenclature.

2.3. Defect analysis

The structures of the as-produced component were examined using a Zeiss Versa 500 Micro-CT system. The Micro-CT was performed using at an accelerating voltage of 120 kV and power of 10 W. The sample was rotated 360° and 1600 projections (at a step of 0.225°) at 1 s exposure time were collected on a charge-coupled device detector. Micro-CT images were then reconstructed using XM Reconstructor software. The Micro-CT 3D raw data was analyzed by an Avizo 8.0 software. Based on the Micro-CT data, pore size and distribution were used to be calculated using the Hildebrand method [25]. The porosity was determined by varying the grayscale threshold to the reconstructed model of CT scan. The threshold value was determined by experience. For samples with the same diameter, the same grayscale was used. Due to the size difference between the 1.2 mm and 7 mm samples, different micro-CT parameters were used and produced a voxel size of 1.9397 μm for 1.2 mm samples and 13.4889 μm for 7 mm samples. The same parameters were used for samples with same diameters produced by SLM and EBM.

2.4. Evaluation of mechanical properties

The tensile and fatigue tests were conducted in air at room temperature using an Instron 8872 machine. Specimens for tensile tests (ASTM: E8) and fatigue tests (ASTM: E466) are shown in Fig. 2(b), the gauge length were 20 mm. Uniaxial tensile tests were carried out at a strain rate of $1.3 \times 10^{-4} \text{ s}^{-1}$. The fatigue tests were performed at stress ratios R of 0.1, a frequency of 30 Hz and a sinusoidal cyclic waveform loading.

3. Results

3.1. Surface characteristics

The surface of the 7 mm diameter rods are shown in Fig. 3. It is apparent that the cylindrical surface of SLM samples is smoother compared to that of EBM samples. Along the vertical orientation, the surface of SLM samples are characterized by a wavy appearance, which is oriented parallel to the build direction. Along the horizontal orientation, the curved surfaces of both SLM and EBM samples are formed by

consecutive stair steps. Both samples contain partly melted powder particles adhered to the surface. It is apparent that the size of the partly melted powders that are adhered to the surface is significantly higher for EBM compared with SLM. This is true for both orientations.

3.2. Defect analysis

The CT-scan results (Fig. 4) indicate that the pores are the main defect in as-built EBM and SLM samples. The pores in EBM samples are almost spherical whereas they tend to be irregular in the SLM samples. There is no significant variation of porosity with part size for both SLM and EBM samples. Compared to EBM samples ($\sim 5 \times 10^{-5}$), the porosity of SLM samples ($\sim 3 \times 10^{-3}$) is much higher. In addition, for the larger SLM sample (Fig. 4g) the pores tend to group along in parallel regions, ~ 3 mm apart.

3.3. Phase analysis

Fig. 5 shows the XRD pattern of Ti–6Al–4V samples fabricated by EBM and SLM. In both samples, most of the peaks can be indexed as the α/α' phase. As α and α' have the same hexagonal close-packed (hcp) structure and a similar lattice parameter, it is difficult to differentiate the two phases. However, in EBM samples, the body centered cubic (bcc) β phase peak is also present. The α/α' peaks of SLM samples are wider than those of EBM samples, indicating that SLM samples possess finer α phase than EBM samples. Ti–6Al–4V is a typical ($\alpha + \beta$) alloy, however, the peaks of β phase are not observed in SLM samples. In order to verify the existence of β phase, the microstructure of both SLM and EBM samples was analyzed using TEM (Fig. 6). In the EBM samples, both bcc and hcp diffraction spots were observed, which confirms the existence of both the α and β phases. The thickness of the β phase is much thinner ($\sim 0.30 \mu\text{m}$) than that of α phase. Further, the volume fraction of β is quite low (Fig. 6). In SLM samples however, only the typical hcp diffraction spots are present, and the structure is the fine α' martensite.

3.4. Microstructure

3.4.1. Influence of part size

Fig. 7 shows the optical micrographs of different sized SLM and EBM samples. The EBM samples are composed of α lamellas and a small amount of β phase, while SLM samples contain much thinner and longer martensitic laths. The thickness of lamellas in EBM samples decreases with the part size. However, for SLM samples, the part size appears to have little effect on the microstructure. Based on both optical and TEM observations, the thickness of α/α' lamellae in EBM and SLM samples was measured and is summarized in Fig. 8. These results show that for all part sizes, the α lamellas in EBM samples are coarser than the α' martensitic laths in the SLM samples. In addition, the thickness of the laths decreases with part size for both SLM and EBM samples, however the change in EBM samples ($1.4 \mu\text{m}$ to $0.4 \mu\text{m}$, $\Delta\lambda \approx 1 \mu\text{m}$) is greater than that in the SLM samples ($0.4 \mu\text{m}$ to $0.1 \mu\text{m}$, $\Delta\lambda \approx 0.3 \mu\text{m}$).

Table 2

The condition and abbreviation of samples fabricated by EBM and SLM process.

Method	EBM		SLM		HIP (SLM&EBM)
Orientation	Vertical-V	Horizontal-H	Vertical-V	Horizontal-H	Vertical-V
Size (mm)	$\Phi 7, \Phi 4, \Phi 1.2$	$\Phi 7, \Phi 4, \Phi 1.2$	$\Phi 7, \Phi 4, \Phi 1.2$	$\Phi 7, \Phi 4, \Phi 1.2$	$\Phi 7$
Abbreviation	EBM-V7	EBM-H7	SLM-V7	SLM-H7	H-SLM-V7
	EBM-V4	EBM-H4	SLM-V4	SLM-H4	&
	EBM-V1.2	EBM-H1.2	SLM-V1.2	SLM-H1.2	H-EBM-V7

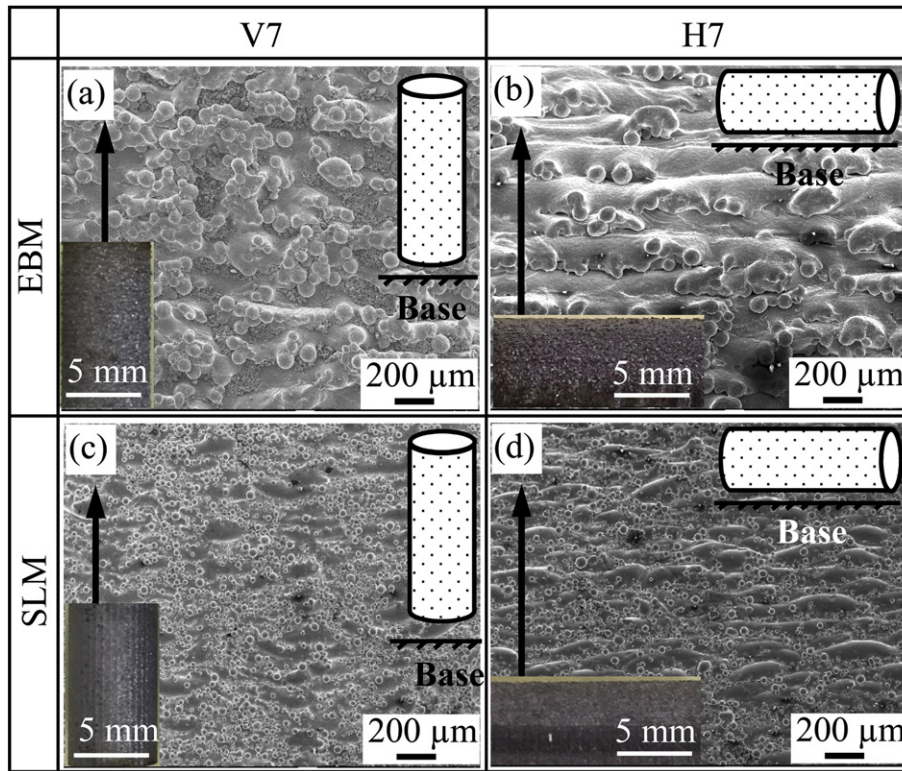


Fig. 3. Cylindrical surface morphology of samples fabricated by EBM and SLM process along different orientation. (a) EBM-V7, (b) EBM-H7, (c) SLM-V7, and (d) SLM-H7. Arrows show the build direction. The inset images are the macro observations of the cylindrical surface.

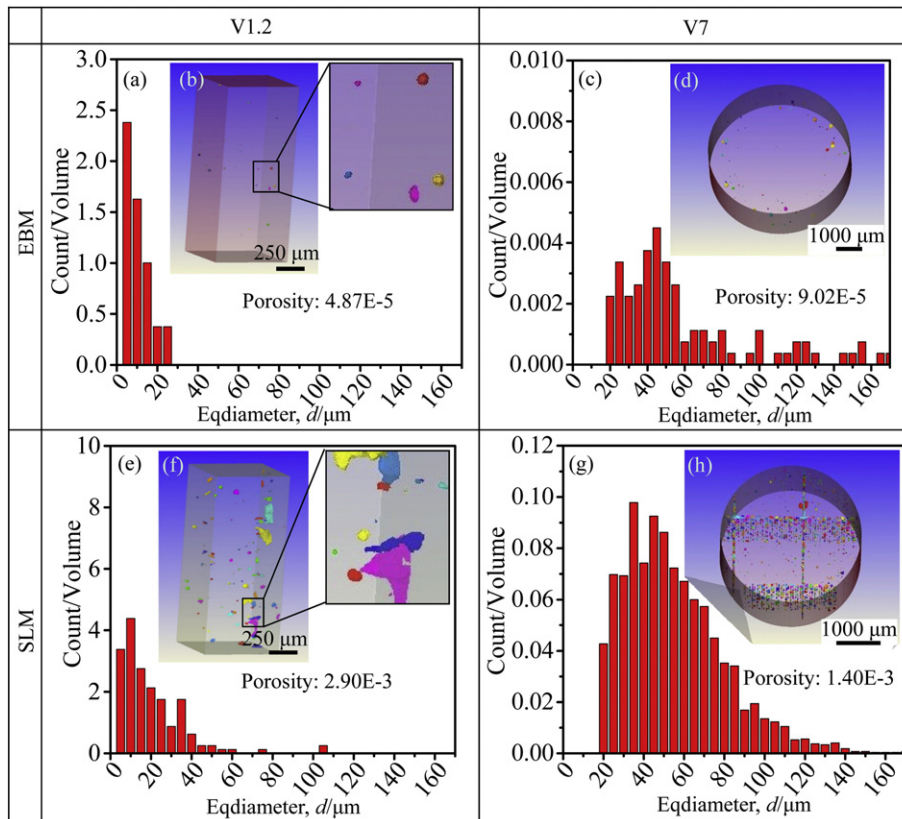


Fig. 4. CT-scan results ((b), (d), (f) and (h)) and pore distribution ((a), (c), (e), and (g)) of as-fabricated samples with different part size fabricated by EBM and SLM process. (a) and (b) EBM-V1.2, (c) and (d) EBM-V7, (e) and (f) SLM-V1.2, and (g) and (h) SLM-V7.

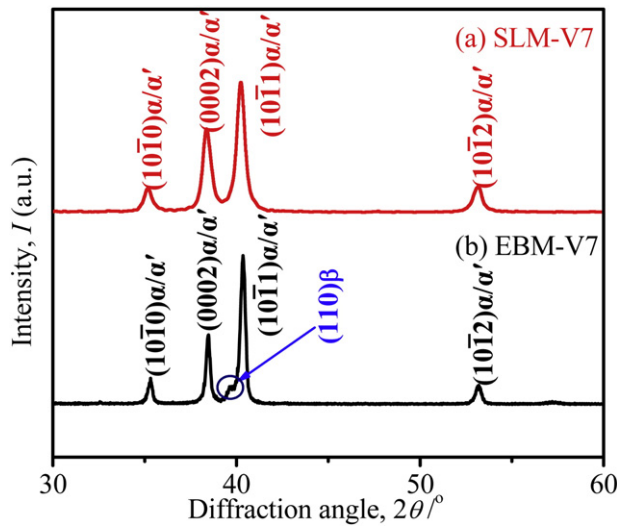


Fig. 5. XRD patterns of Ti-6Al-4V samples with diameter in 7 mm fabricated by SLM and EBM process.

3.4.2. Influence of orientation

Fig. 9 shows the microstructure of as-fabricated Ti-6Al-4V samples with diameter of 7 mm along different orientations. Both SLM and EBM samples in both the vertical and horizontal orientations are dominated by columnar grains which tend to elongate along the build direction. These columnar grains usually grow over many layers. For SLM samples, martensitic needles are obvious within the grains. For EBM samples, the microstructure is mainly composed of α phase and a small amount of β within the prior β columnar grains. Grain boundary α phase is also present around the boundaries of the columnar prior β grains.

3.4.3. Influence of HIP treatment

The optical microstructure of as-fabricated and HIPed Ti-6Al-4V samples fabricated by EBM and SLM are compared in Fig. 10. It is apparent that the HIP treatment significantly coarsens the α lamellae in both EBM and SLM samples. The coarsened α phase after HIP treatment is attributed to the high heat treatment temperature (1203 K)

which is close to β transus temperature ($1253 \pm 10\text{ K}$) and the subsequent slow cooling (furnace cooling). Compared with EBM samples, the α lamellae in the SLM samples are much thicker and longer. Optical observations show that the as-built SLM samples consist of long and fine α' needles. When heated, α phase is nucleated along the α' boundaries and vanadium atoms are expelled, leading to the formation of β at α phase boundaries [26,27]. Hence during HIP treatment, the long α' needles are transformed into the long and thick α lamellae shown in Fig. 10d.

3.5. Mechanical properties

3.5.1. Tensile properties

Fig. 11 shows the tensile properties of EBM and SLM Ti-6Al-4V samples with different part sizes and orientations. The yield stress (YS) and ultimate tensile strength (UTS) of EBM and SLM samples are comparable with or even better than those of wrought alloys (YS—850 MPa; UTS—930 MPa) given in the ASM Handbook [28]. The strength of SLM samples is higher than those of EBM samples whereas the ductility is much lower. For EBM samples, the strength increases, while elongation and area reduction decreases with decreasing part size. Such tendency is more evident when the part size is lower than 4 mm. Unlike EBM samples, the strength of SLM samples decreases with the part size and as does the elongation. For both EBM and SLM samples, the strength and ductility along vertical orientation is slightly higher than those along horizontal orientation. HIP treatment significantly improves the ductility but decrease the strength of both SLM and EBM samples. The drop in strength is much greater in SLM samples than EBM.

3.5.2. Fatigue performance

Fig. 12 shows the S-N curves illustrating the fatigue behavior of as-fabricated and HIPed Ti-6Al-4V samples. Due to the pores contained in as-built samples, the fatigue limit of the SLM/EBM samples are lower than that of cast ($\sim 450\text{ MPa}$) [29] and annealed alloys ($\sim 500\text{ MPa}$) [30]. EBM samples, having a lower level of porosity, have a slightly higher fatigue limit than SLM parts. During the HIP treatment, most of pores in EBM and SLM samples are closed and the fatigue strength is significantly improved to above 550 MPa.

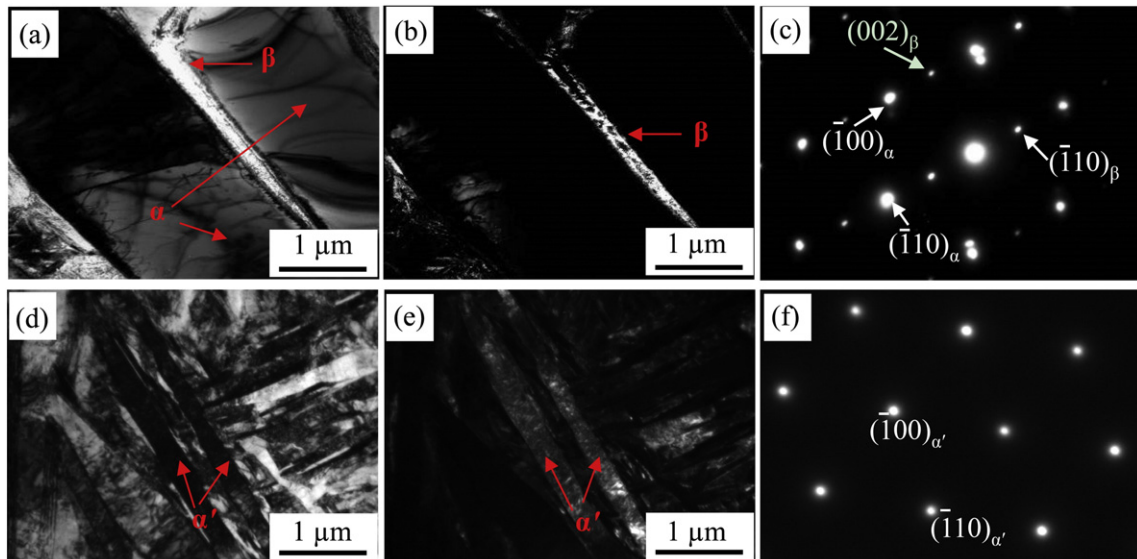


Fig. 6. Bright field images ((a) and (d)), dark field images ((b) and (e)), and diffraction pattern ((c) and (f)) of samples with diameter of 7 mm fabricated by SLM and EBM process: (a), (b), and (c) EBM-V7; (d), (e), and (f) SLM-V7. The beam incident direction is $[001]_{\alpha}$.

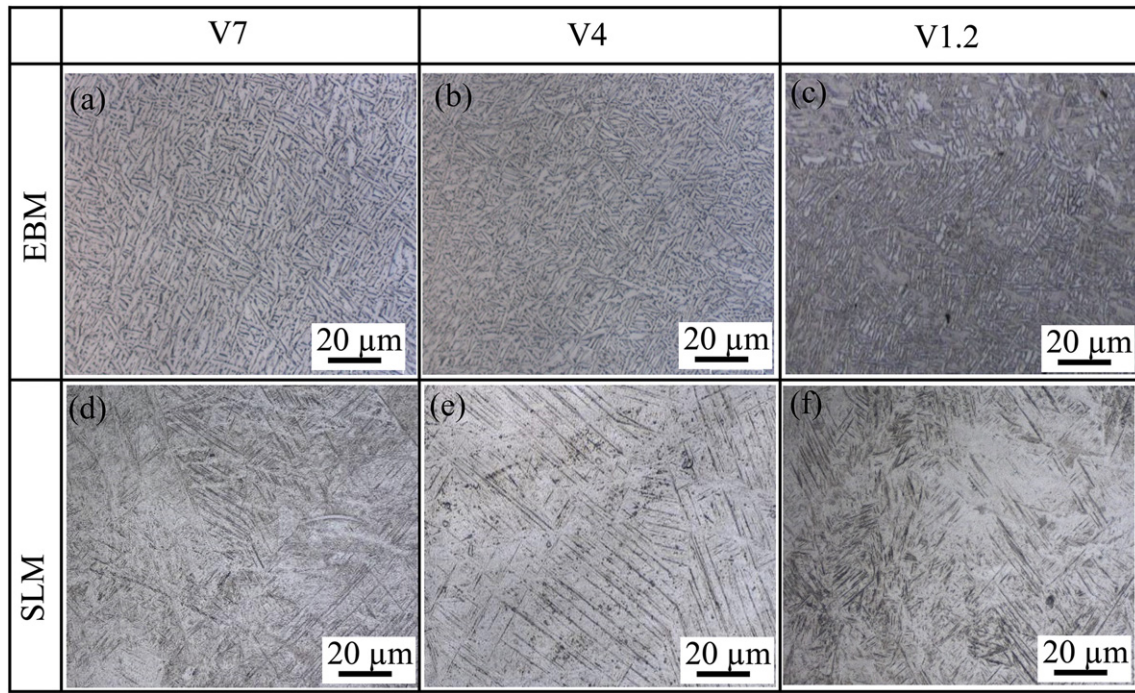


Fig. 7. Optical microstructures of as-fabricated EBM and SLM samples with different part size along vertical orientation. (a) EBM-V7, (b) EBM-V4, (c) EBM-V1.2, (d) SLM-V7, (e) SLM-V4, and (f) SLM-V1.2.

3.6. Morphology of fracture surfaces

3.6.1. Tensile fracture

Representative tensile fractography of SLM and EBM Ti-6Al-4V samples with different orientation is shown in Fig. 13. The fracture surface of EBM samples along vertical orientation is characterized by typical ductile dimple tearing resulting from the coalescence of micro voids (Fig. 13a). A significant population of fine and deeper dimples at the tensile fracture surface indicates the extent of plastic deformation. Fracture surface of EBM samples along horizontal orientation (Fig. 13b) exhibits a mixed mode of ductile and brittle (quasi-cleavage) fracture showing predominantly cleavage facets and some transgranular fracture. For the SLM samples, the macro fracture surface is flat and the necking is not so obvious compared to EBM samples (Fig. 13c, d). However, the micro fracture surfaces are characterized by shallow dimples, cleavage facets, and lots of opened-up pores. These indicate a mixed mode of ductile and brittle fracture. The amount of

porosity on the fracture surface is greater in the SLM samples than EBM. There is also a difference in the amount of porosity between the horizontal and vertical samples, with the horizontal orientation having a great number of pores.

3.6.2. Fatigue fracture

Fig. 14 shows the fatigue fracture surfaces of as-fabricated and HIPed Ti-6Al-4V samples fabricated by SLM and EBM process. For both as-built EBM and SLM samples (Fig. 14a and b), the cracks initiate from an internal pores in the sub-surface and propagated radically outwards. This indicates that the pores are the dominant feature which controls the fatigue performance. For HIPed EBM and SLM samples, the cracks initiated on the surface of samples, which are generally attributed to surface defects or the classical slip-band intrusion/extrusion mechanism (Fig. 14c and d).

4. Discussions

The different characteristics of the microstructure, porosity, surface finish and mechanical properties presented above between EBM and SLM samples may be a result of the different principles of EBM and SLM systems. SLM systems use mirror deflection systems to scan the laser beam, which permit accurate laser beam scanning up to ~15 m/s [6]. EBM systems, in contrast, resemble classical electron optical configurations such as scanning or transmission electron microscopes. Although conceptually similar to SLM, EBM systems generate a high energy electron beam in a standard electron gun configuration operating at 60 kV accelerating potential. Beam currents in the tens of milliamperes are used to create larger beam energies than those used in SLM. The electron beam is also used to preheat the powder prior to the melt scanning, which reduces the thermal gradients and therefore cooling rate. In SLM, only the substrate is heated; in this work to 473 K. Due to the lower substrate temperatures, SLM is characterized by high temperature gradients, which results in rapid solidification causing the build-up of thermal stresses, and the presence of non-equilibrium phases [31]. In addition, other parameters, such as the powder size, scan spacing of SLM and EBM system are also different.

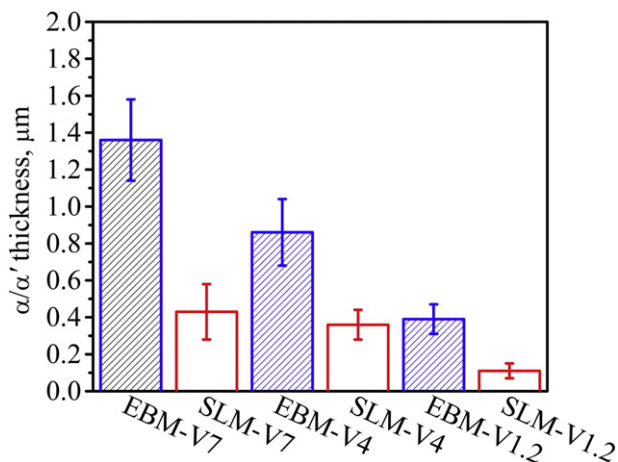


Fig. 8. The α/α' thickness in as-fabricated EBM and SLM samples with different part size.

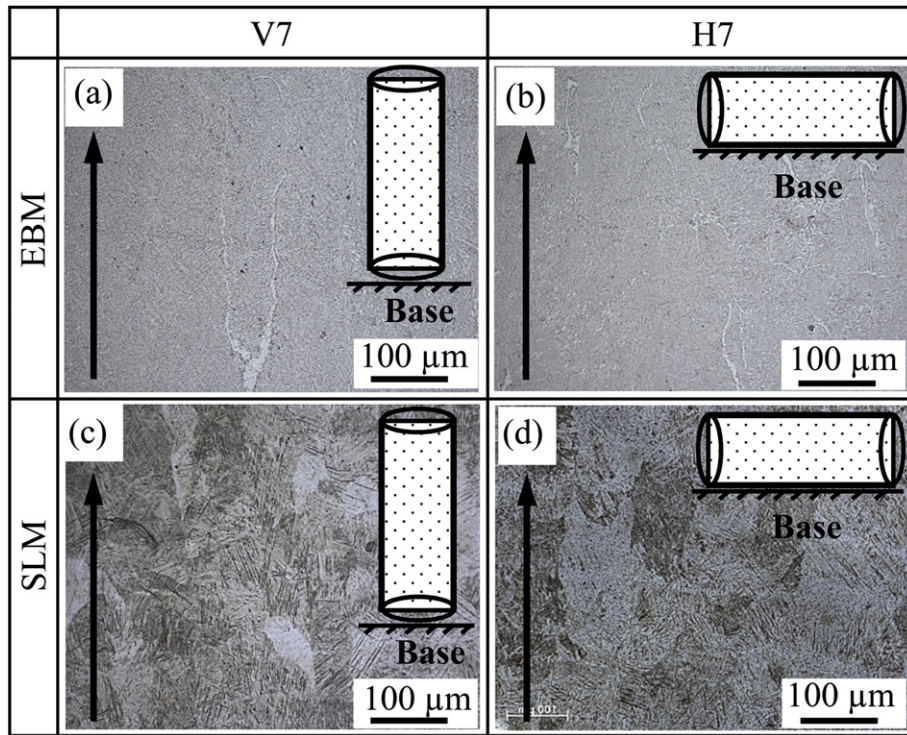


Fig. 9. Optical microstructures of as-fabricated Ti–6Al–4 V samples along different orientations: (a) EBM–V7, (b) EBM–H7, (c) SLM–V7, and (d) SLM–H7. The arrows show the build direction. In (a) and (c), the orientation is parallel to the build direction; in (b) and (d) the orientation is perpendicular to the build direction.

All these physical differences between the SLM and EBM systems results in different characteristics of SLM and EBM samples.

4.1. Surface characteristics

For SLM samples, the numbers of semi-molten particles adhering on the surface is due to the attenuation of the laser irradiation on the

boundary of model. The smooth surfaces for SLM samples are mainly due to the lower energy input, narrower scan spacing, smaller powder size, and slower scan speeds. For EBM samples, the powder is held at a much higher temperature in EBM and therefore heat diffusing out from the melt pool is more likely to cause the surrounding powder to be sintered (or partially melted) onto part. In addition, the higher scan speeds and larger scan spacing

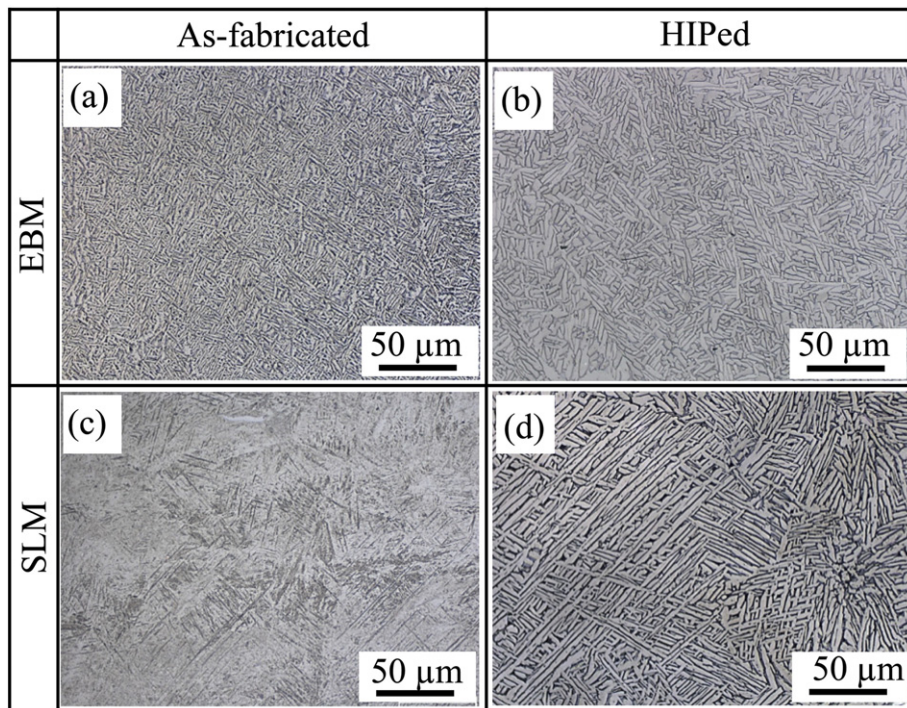


Fig. 10. Optical microstructures of as-fabricated ((a) and (c)) and HIPed ((b) and (d)) Ti–6Al–4V samples fabricated by EBM ((a) and (b)) and SLM ((c) and (d)) process.

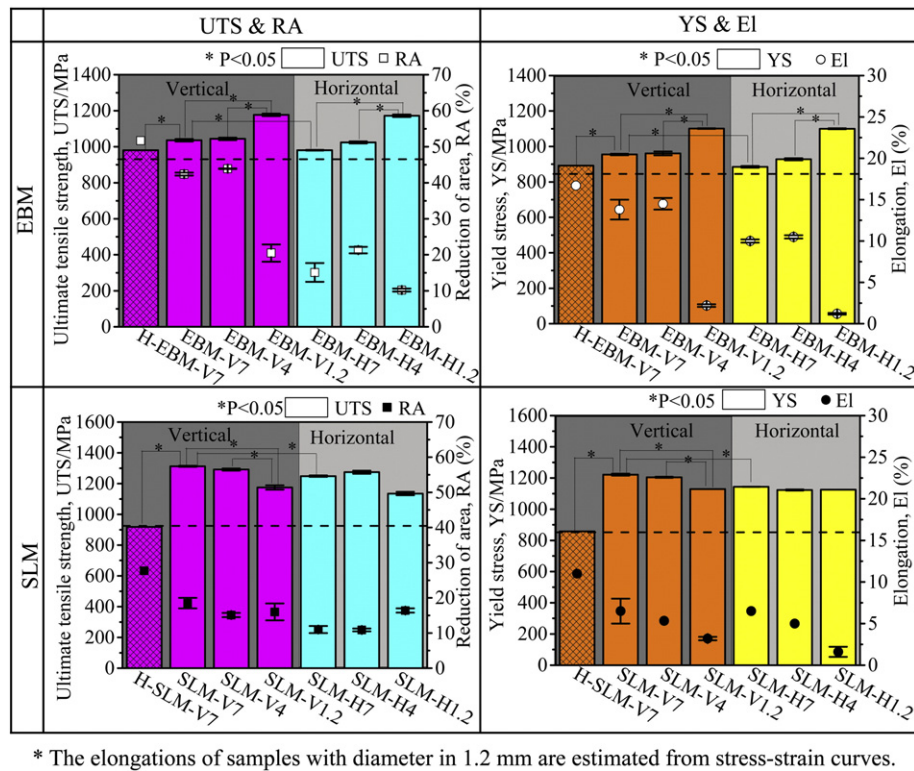


Fig. 11. Tensile properties of Ti-6Al-4V samples with different part size, orientation, and fabrication methods. The dash lines are corresponding to the properties of wrought alloys.

for the EBM system makes the EBM process faster at the expense of poor surface finish.

4.2. Pore characteristics

It is evident that porosity forms during both EBM and SLM processing (Fig 4), and the pore distribution and pore characteristics are quite different in SLM and EBM samples. The differences of pore distribution in EBM and SLM samples may be due to the different scanning strategies used. For EBM samples, electron beam move across the part bed in a raster scan pattern, and the pores generally formed near the turning point on the edge of the samples (Figs. 4, 15a). In contrast, the SLM samples were produced in 3 mm stripes and it is apparent that the majority of the pores in the large sample sit at the interface between the stripes (Fig. 15b). These

results indicate that the distribution of pores is closely related to the scanning strategies of SLM system. For the smaller 1.2 mm diameter sample (Fig 4c), the part was completed in one “stripe” and therefore the location of the porosity is more random. The different pore morphology between EBM and SLM samples may be attributed to their different beam energy. For SLM samples, the formation of flat or irregular-shaped pores is due to incomplete homologous melting and solidification, especially for the part with the smallest size. For EBM, the input beam energy is much higher than that of SLM and the whole process is maintained at a much higher temperature, which is beneficial for complete melting of the powder. Thus, the EBM samples contained fewer and smaller pores and the detected spherical pores are mainly due to the gas bubbles (or spherical voids) in the atomized powder formed during the rapid solidification process and evaporation of elements with low boiling point [32].

4.3. Microstructure and phase composition

Both EBM and SLM differs markedly from traditional solidification as a consequence of undercooling at the solid/liquid (melt) interface. The lower substrate temperature (~ 473 K) and increased convective cooling due to the presence of Ar in SLM result in a faster cooling rate than that experienced during EBM. Since the SLM process has a very high cooling rate (usually greater than 10^4 K/s), the martensitic transformation to α' occurs [22]. The obvious martensitic needles within the grains are generally inclined at about 40° to the build direction [15]. This is due to the special Burgers relation between α/α' phase and β phase which dictates the α' preferential growth orientation during fast cooling of SLM [27]. In addition, due to the much lower substrate temperature of SLM, the effect of the scanned area on the cooling rate is significantly reduced and the part size has much less effect on the microstructure (Fig. 8).

For the EBM technique, the substrate is maintained at a higher temperature of 673 ~ 823 K and the building is processed in vacuum. As such, the cooling rate of powder bed is much lower than that of SLM and therefore the EBM samples mainly consist of α instead of α' . With increasing part size, the surface area of sample increases, meaning

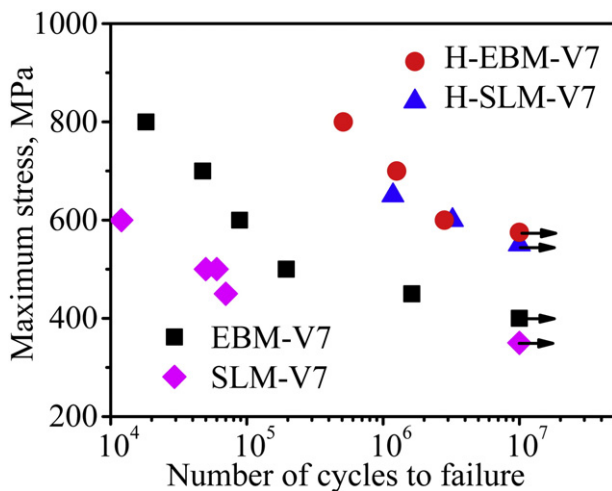


Fig. 12. S-N curves of as-fabricated and HIPed Ti-6Al-4V samples fabricated by EBM and SLM process.

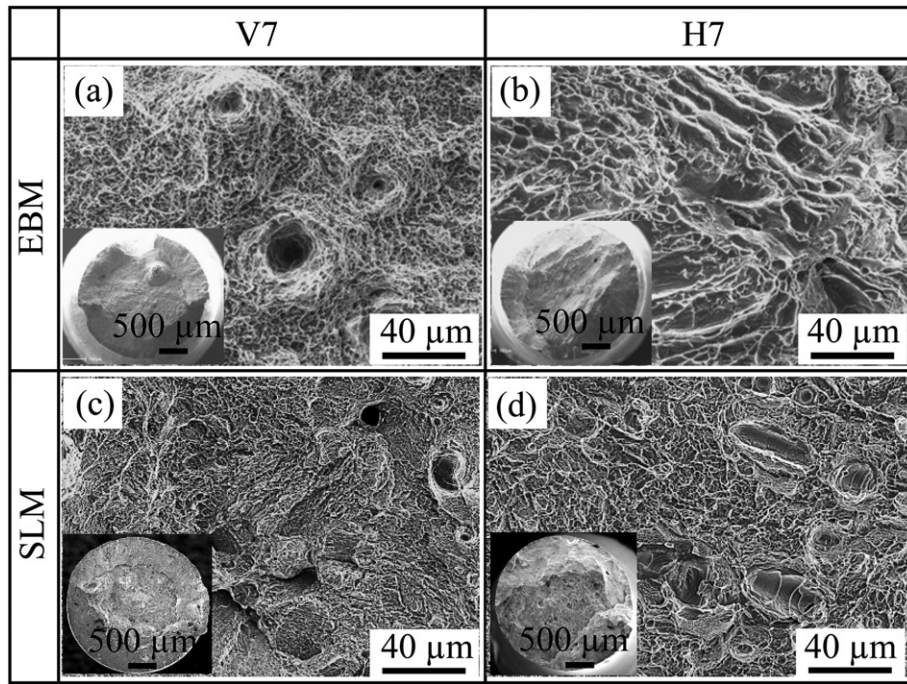


Fig. 13. Tensile fracture surface of as-fabricated EBM and SLM samples along different orientations. (a) EBM-V7, (b) EBM-H7, (c) SLM-V7, and (d) SLM-H7.

more heat is generated during the building process. This in turn results in a slower cooling rate, resulting in coarser α lamellas.

4.4. Mechanical properties

The mechanical properties of bulk solids are controlled by their microstructures. Thus the above difference in morphology results in different mechanical properties of SLM and EBM samples.

4.4.1. α' martensite

Due to much higher cooling rate during building, the SLM samples mainly contained α' martensite instead of the coarser α phase apparent in the EBM parts. This non-equilibrium, fine α' martensitic phase is characterized by high strength, but low ductility and therefore imparts these properties on the parts. In contrast, the α phase is well known to be weaker, but more ductile than α' .

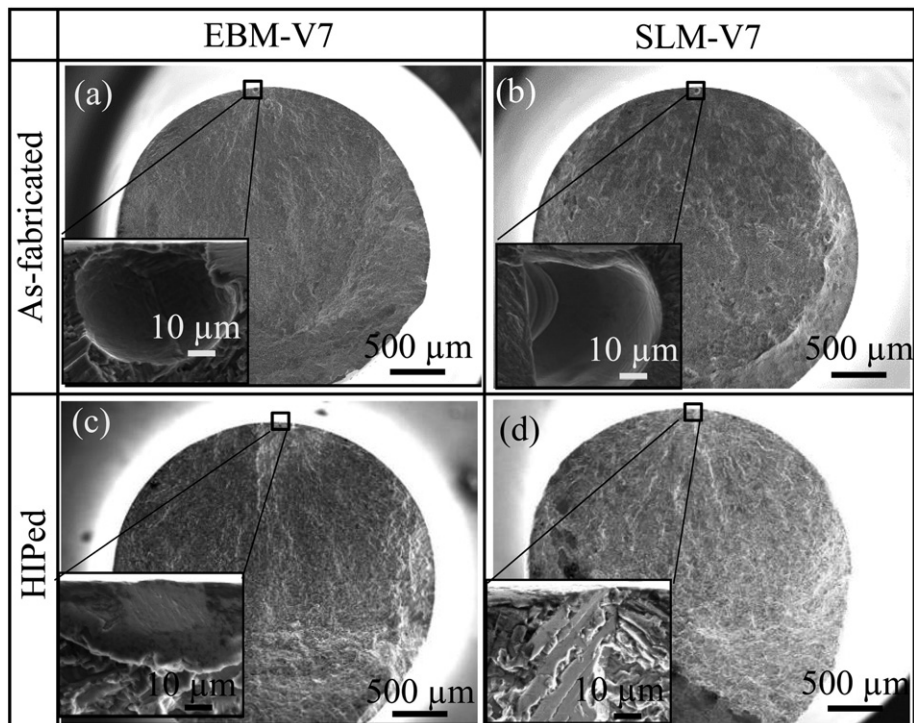


Fig. 14. Fatigue surfaces of as-fabricated and HIPed Ti-6Al-4V samples fabricated by EBM and SLM process. (a) EBM-V7, (b) SLM-V7, (c) H-EBM-V7, and (d) H-SLM-V7.

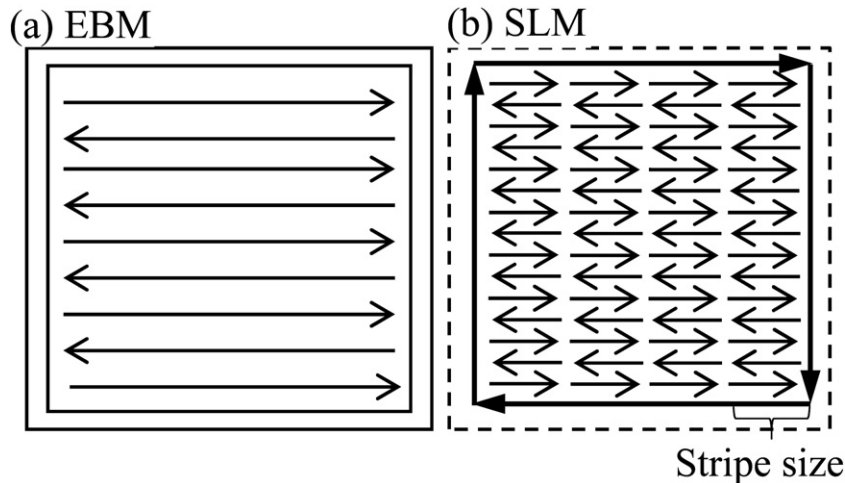


Fig. 15. EBM (a) and SLM (b) scanning strategies.

4.4.2. Variation of tensile properties with part size

For EBM samples, the tensile properties are significantly affected by the part size, especially for samples with smaller size (~4 mm), with the strength decreasing and the ductility increasing with increasing part size. This can be attributed to two effects. The first is related to the change in thickness of α lamellas as discussed in Section 4.3. For two phase titanium alloys, it is well known that the relationship between strength and α thickness follow the Hall–Petch relation [10,33,34]:

$$\sigma_{ys} = \sigma_0 + K_y d^{-1/2} \quad (1)$$

where σ_0 is the yield strength of single crystal, K_y is the strengthening co-efficient, and d is the thickness of lamellas. Therefore, it is expected that the observed increase in the α lamellae thickness will cause a decrease in strength. The second effect may be a decrease in dislocation density within α lamellas at the larger part, which has been shown to occur as a result of the variation in cooling rate [35]. This decrease in dislocation density will also act to soften the material as the part size increases.

The tensile strength and ductility of SLM samples is also affected by the part size, especially for the smaller samples (~4 mm) (Fig. 11). Since there is much less variation in the scale of the microstructure (Fig. 8) for the SLM samples, a more constant property set is expected. The decrease in the smallest sized samples may be related to the porosity. Although the average pore size decreases from ~50 μm to ~20 μm (Fig. 4), the pores size relative to the cross-sectional area become larger. Hence, for sample with small diameters, pores will occupy a significantly higher proportion of the cross-sectional area and therefore tend to be more deleterious to the strength. Although the same should be true for EBM sample, the lower overall porosity coupled with the changes in the size of the α lamellae tends to mask any affect.

4.4.3. Variation of tensile properties along different orientations

The tensile strength and ductility of samples in vertical orientation are superior to those in horizontal direction for both EBM and SLM samples (Fig. 11). Such difference on tensile properties is mainly attributed to the difference between the prior β columnar grain orientation and the tensile axis direction. The prior β columnar grain orientations in both EBM and SLM samples are parallel to the build direction of the samples. This means that for the samples in the vertical orientation, the tensile axis direction is parallel to the prior β columnar grain orientation. During deformation, a large amount of grains will contribute to necking of the samples and therefore the samples can withstand a larger degree of deformation, which leads to higher ductility, especially the reduction of area. For the samples in horizontal orientation, the tensile axis direction is perpendicular to the orientation of the prior β grains.

When load is applied to the samples, large stresses maybe concentrated near the grain boundaries. Dimples easily initiate at the α/β interface near the grain boundary and transgranular fracture occurs along the grain boundary, resulting in lower ductility.

4.4.4. Effect of HIP on the fatigue properties of SLM and EBM samples

The pores are the main defect in SLM and EBM samples. These pores seem to have less negative effect on tensile properties of SLM and EBM samples, as evidenced by high strength of SLM samples with higher porosity as compared to EBM samples. However, fatigue cracks can easily originate around these pores and significantly decrease the fatigue strength of both EBM and SLM samples. HIP treatment can close most of the pores contained in EBM and SLM samples. Although due to coarsening of α phase during HIP treatment, the tensile strength decrease and ductility increase, (especially true for the SLM samples), their fatigue strength are significantly improved to above 550 MPa.

5. Conclusions

In this study, the microstructures and mechanical properties of EBM and SLM Ti–6Al–4V parts with different part size and orientation were systematically investigated and compared. The following conclusions were drawn:

- (1) The EBM and SLM Ti–6Al–4V samples mainly consisted of $\alpha + \beta$ and α' phase, respectively. Due to much finer α' martensite in SLM sample, the tensile strength of SLM samples is significantly greater than that of EBM samples; whereas the ductility is lower.
- (2) Pore defects were present in both EBM and SLM samples. Compared to EBM samples, the porosities of SLM samples are much higher. Due to the different scanning strategies, the pores generally formed near the turning point on the edge of the EBM samples whereas they were at the interface between the scanning stripes for SLM samples.
- (3) The thickness of the α lamellas decreases with decrease in part size during EBM fabrication as a result of the faster cooling rate. This resulted in an improvement in strength and decrease in ductility. For SLM samples, the part size has almost no effect on the mechanical properties.
- (4) EBM and SLM Ti–6Al–4V samples in the vertical orientation possess higher ultimate tensile strength, yield strength, and better ductility than those in the horizontal orientation. The difference is mainly attributed to a change in tensile axial direction relative to the orientation of the prior β grains.
- (5) Due to the pores contained in the as-fabricated samples, the fatigue strength of both EBM and SLM Ti–6Al–4V samples is

lower than those of cast and annealed alloys. After HIP treatment, most of pores in EBM and SLM samples are closed and the fatigue strength is significantly improved to above 550 MPa.

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References

- [1] W. Frazier, Metal additive manufacturing: a review, *J. Mater. Eng. Perform.* 23 (2014) 1917–1928.
- [2] M. Koike, P. Greer, K. Owen, G. Lilly, L.E. Murr, S.M. Gaytan, et al., Evaluation of titanium alloys fabricated using rapid prototyping technologies—electron beam melting and laser beam melting, *Materials* 4 (2011) 1776–1792.
- [3] C. Yan, L. Hao, A. Hussein, P. Young, D. Raymont, Advanced lightweight 316L stainless steel cellular lattice structures fabricated via selective laser melting, *Mater. Des.* 55 (2014) 533–541.
- [4] T. Koutsoukis, S. Zinelis, G. Eliades, K. Al-Wazzan, M.A. Rifaai, Y.S. Al Jabbari, Selective laser melting technique of Co–Cr dental alloys: a review of structure and properties and comparative analysis with other available techniques, *J. Prosthodont.* 24 (2015) 303–312.
- [5] X.P. Li, X.J. Wang, M. Saunders, A. Suvorova, L.C. Zhang, Y.J. Liu, et al., A selective laser melting and solution heat treatment refined Al–12Si alloy with a controllable ultra-fine eutectic microstructure and 25% tensile ductility, *Acta Mater.* 95 (2015) 74–82.
- [6] L.E. Murr, *Handbook of Materials Structures, Properties, Processing and Performance*, Springer International Publishing, Switzerland, 2015.
- [7] I. Yadroitsev, P. Krakhmalev, I. Yadroitsava, Selective laser melting of Ti6Al4V alloy for biomedical applications: temperature monitoring and microstructural evolution, *J. Alloys Compd.* 583 (2014) 404–409.
- [8] E. Sallica-Leva, A.L. Jardini, J.B. Fogagnolo, Microstructure and mechanical behavior of porous Ti–6Al–4V parts obtained by selective laser melting, *J. Mech. Behav. Biomed. Mater.* 26 (2013) 98–108.
- [9] S.S. Al-Bermani, M.L. Blackmore, W. Zhang, I. Todd, The origin of microstructural diversity, texture, and mechanical properties in electron beam melted Ti–6Al–4V, *Metall. Mater. Trans. A* 41 (2010) 3422–3434.
- [10] L.E. Murr, E.V. Esquivel, S.A. Quinones, S.M. Gaytan, M.I. Lopez, E.Y. Martinez, et al., Microstructures and mechanical properties of electron beam-rapid manufactured Ti–6Al–4V biomedical prototypes compared to wrought Ti–6Al–4V, *Mater. Charact.* 60 (2009) 96–105.
- [11] L.E. Murr, S.A. Quinones, S.M. Gaytan, M.I. Lopez, A. Rodela, E.Y. Martinez, et al., Microstructure and mechanical behavior of Ti–6Al–4V produced by rapid-layer manufacturing, for biomedical applications, *J. Mech. Behav. Biomed. Mater.* 2 (2009) 20–32.
- [12] N. Hrabec, T. Quinn, Effects of processing on microstructure and mechanical properties of a titanium alloy (Ti–6Al–4V) fabricated using electron beam melting (EBM), part 2: energy input, orientation, and location, *Mater. Sci. Eng. A* 573 (2013) 271–277.
- [13] N. Hrabec, T. Quinn, Effects of processing on microstructure and mechanical properties of a titanium alloy (Ti–6Al–4V) fabricated using electron beam melting (EBM), part 1: distance from build plate and part size, *Mater. Sci. Eng. A* 573 (2013) 264–270.
- [14] J. Karlsson, A. Snis, H. Engqvist, J. Lausmaa, Characterization and comparison of materials produced by electron beam melting (EBM) of two different Ti–6Al–4V powder fractions, *J. Mater. Process. Technol.* 213 (2013) 2109–2118.
- [15] C. Qiu, N.J.E. Adkins, M.M. Attallah, Microstructure and tensile properties of selectively laser-melted and of HIPed laser-melted Ti–6Al–4V, *Mater. Sci. Eng. A* 578 (2013) 230–239.
- [16] B. Song, S.J. Dong, B.C. Zhang, H.L. Liao, C. Coddet, Effects of processing parameters on microstructure and mechanical property of selective laser melted Ti6Al4V, *Mater. Des.* 35 (2012) 120–125.
- [17] P. Edwards, M. Ramulu, Fatigue performance evaluation of selective laser melted Ti–6Al–4V, *Mater. Sci. Eng. A* 598 (2014) 327–337.
- [18] S. Lenders, M. Thoenes, A. Riemer, T. Niendorf, T. Troester, H.A. Richard, et al., On the mechanical behaviour of titanium alloy Ti6Al4V manufactured by selective laser melting: fatigue resistance and crack growth performance, *Int. J. Fatigue* 48 (2013) 300–307.
- [19] V. Cain, L. Thijs, J. Van Humbeeck, B. Van Hooreweder, R. Knutsen, Crack propagation and fracture toughness of Ti6Al4V alloy produced by selective laser melting, *Addit. Manuf.* 5 (2015) 68–76.
- [20] H.K. Rafi, T. Starr, B. Stucker, A comparison of the tensile, fatigue, and fracture behavior of Ti–6Al–4V and 15–5 PH stainless steel parts made by selective laser melting, *Int. J. Adv. Manuf. Technol.* 69 (2013) 1299–1309.
- [21] L. Ladani, J. Razmi, C.S. Farhan, Mechanical anisotropy and strain rate dependency behavior of Ti6Al4V produced using E-Beam Additive Fabrication, *J. Eng. Mater. Technol.* 136 (2014) 031006–1–7.
- [22] H.K. Rafi, N.V. Karthik, H. Gong, T.L. Starr, B.E. Stucker, Microstructures and mechanical properties of Ti6Al4V parts fabricated by selective laser melting and electron beam melting, *J. Mater. Eng. Perform.* 22 (2013) 3872–3883.
- [23] A. Simchi, Direct laser sintering of metal powders: mechanism, kinetics and microstructural features, *Mater. Sci. Eng. A* 428 (2006) 148–158.
- [24] R. Boyer, G. Welsch, E.W. Collings, *Materials Properties Handbook: Titanium Alloys*, ASM International, Materials Park, OH, 1994.
- [25] T. Hildebrand, P. Rüegsegger, A new method for the model-independent assessment of thickness in three-dimensional images, *J. Microscopy* 185 (1997) 67–75.
- [26] F.X. Gil Mur, D. Rodríguez, J.A. Planell, Influence of tempering temperature and time on the α' -Ti–6Al–4V martensite, *J. Alloys Compd.* 234 (1996) 287–289.
- [27] B. Vrancken, L. Thijs, J.-P. Kruth, J. Van Humbeeck, Heat treatment of Ti6Al4V produced by selective laser melting: microstructure and mechanical properties, *J. Alloys Compd.* 541 (2012) 177–185.
- [28] S. Lampman, Wrought titanium and titanium alloys, in: K.M. Zwilsky (Ed.), *ASM Handbook Volume 2: Properties and Selection Nonferrous Alloys and Special-Purpose Materials*, ASM International, OH 1993, pp. 1782–1886.
- [29] *Metallic materials properties development and standardization (MMPDS)*, Washington, D.C.: U.S. Department of Transportation Federal Aviation Administration; 2003. (5–51–91).
- [30] G. Lütjering, Influence of processing on microstructure and mechanical properties of (α + β) titanium alloys, *Mater. Sci. Eng. A* 243 (1998) 32–45.
- [31] L. Thijs, F. Verhaeghe, T. Craeghs, J.V. Humbeeck, J.-P. Kruth, A study of the microstructural evolution during selective laser melting of Ti–6Al–4V, *Acta Mater.* 58 (2010) 3303–3312.
- [32] S.M. Gaytan, L.E. Murr, F. Medina, E. Martinez, M.I. Lopez, R.B. Wicker, Advanced metal powder based manufacturing of complex components by electron beam melting, *Mater. Technol.* (2009).
- [33] C. Leyens, M. Peters, Z. Chen, *Titanium and Titanium Alloys*, Chemical Industry Press, Beijing, 2005.
- [34] H. Shi, X. Ren, *Mechanical Properties of Materials*, Peking University Press, Beijing, 2010.
- [35] Z. Wang, J. Zhang, S. Li, W. Hou, Y. Hao, R. Yang, The effect of part size on the microstructure and mechanical properties of EBM Ti–6Al–4V alloys, *Rare Metal Mater. Eng.* 161–4 (2014).